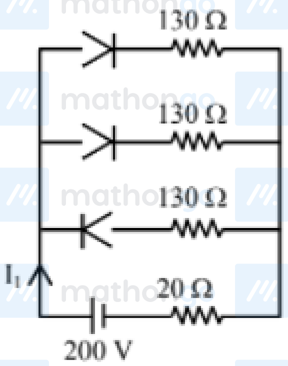
Q1 2021 (01 Sep Shift 2)

In the given figure, each diode has a forward bias resistance of 30Ω and infinite resistance in reverse bias. The current I_1 will be :



(1) 3.75 A

(2) 2.35 A

(3) 2 A

(4) 2.73 A

Q2 2021 (31 Aug Shift 2)

Statement-I :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load R_L

Statement-II :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with R_L.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is true but Statement II is false

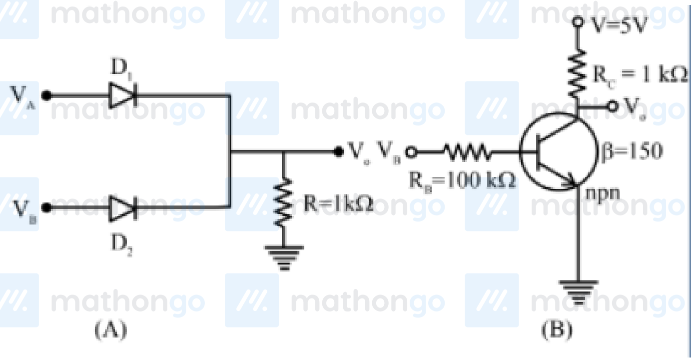
(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are false

(4) Both Statement I and Statement II are true

Q3 2021 (31 Aug Shift 2)

If V_A and V_B are the input voltages (either 5 V or 0 V) and V_o is the output voltage then the two gates represented in the following circuit (A) and (B) are:-



(1) AND and OR Gate

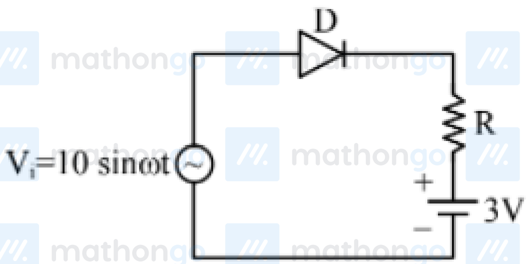
(2) OR and NOT Gate

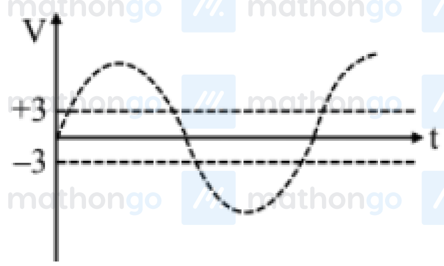
(3) NAND and NOR Gate

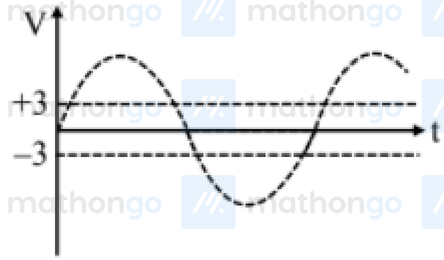
(4) AND and NOT Gate

Q4 2021 (31 Aug Shift 1)

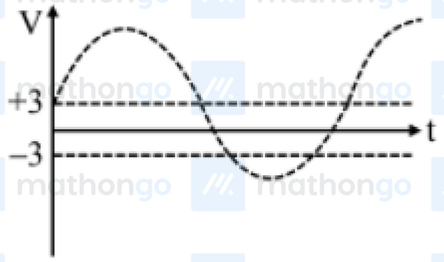
Choose the correct waveform that can represent the voltage across R of the following circuit, assuming the diode is ideal one:



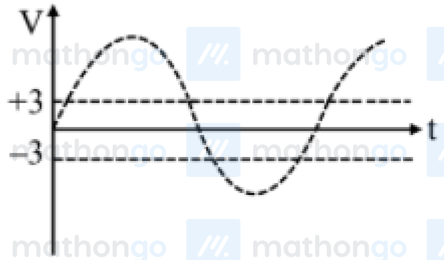
(1)




(2)



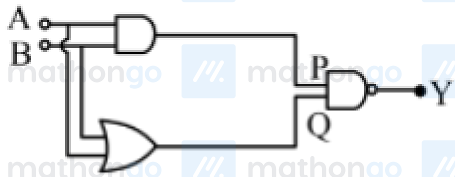
(3)



(4)

Q5 2021 (31 Aug Shift 1)

In the following logic circuit the sequence of th inputs A, B are $(0,0),(0,1),(1,0)$ and $(1,1)$. Th output Y for this sequence will be :



(1) $1,0,1,0$

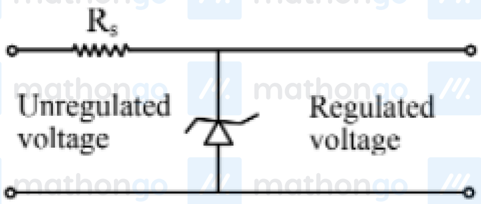
(2) $0,1,0,1$

(3) $1,1,1,0$

(4) $0,0,1,1$

Q6 2021 (27 Aug Shift 2)

A zener diode of power rating 2 W is to be used as a voltage regulator. If the zener diode has a breakdown of 10 V and it has to regulate voltage fluctuated between 6 V and 14 V, the value of R_s for safe operation should be Ω



Q7 2021 (27 Aug Shift 2)

For a transistor α and β are given as $\alpha = \frac{I_C}{I_E}$ and $\beta = \frac{I_C}{I_B}$. Then the correct relation between α and β will be :

(1) $\alpha = \frac{1-\beta}{\beta}$

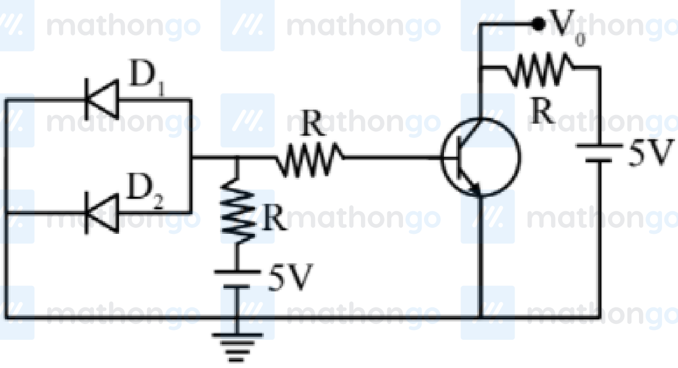
(2) $\beta = \frac{\alpha}{1-\alpha}$

(3) $\alpha\beta = 1$

(4) $\alpha = \frac{\beta}{1-\beta}$

Q8 2021 (27 Aug Shift 1)

A circuit is arranged as shown in figure. The output voltage V_0 is equal to V



Q9 2021 (27 Aug Shift 1)

For a transistor in CE mode to be used as an amplifier, it must be operated in :

(1) Both cut-off and Saturation

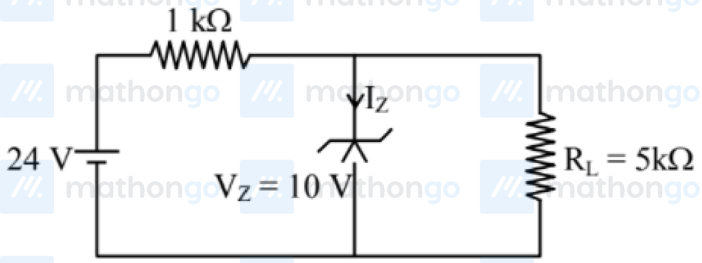
(2) Saturation region only

(3) Cut-off region only

(4) The active region only

Q10 2021 (26 Aug Shift 2)

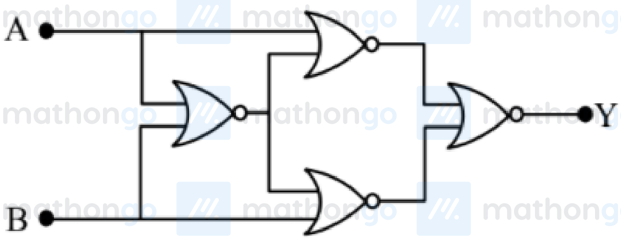
For the given circuit, the power across zener diode is mW



Q11 2021 (26 Aug Shift 2)

Four NOR gates are connected as shown in figure.

The truth table for the given figure is :



(1)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(3)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Q12 2021 (26 Aug Shift 1)

Statement-I : By doping silicon semiconductor with pentavalent material, the electrons density increases.

Statement-II : The n-type semiconductor has net negative charge.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I is true but Statement-II is false.

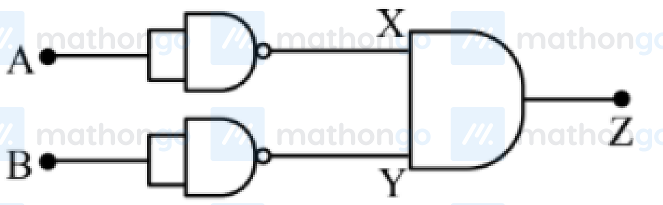
(2) Statement-I is false but Statement-II is true.

(3) Both Statement-I and Statement-II are true.

(4) Both Statement-I and Statement-II are false.

Q13 2021 (26 Aug Shift 1)

Identify the logic operation carried out by the given circuit :-



(1) OR

(2) AND

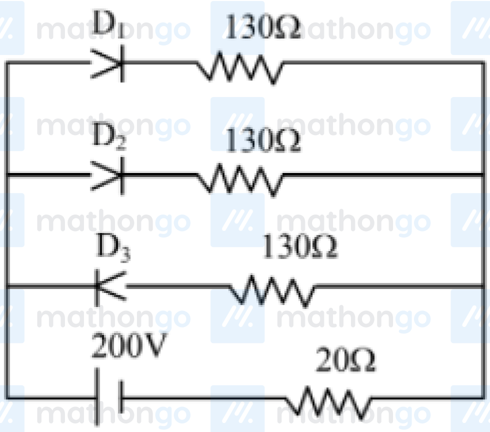
(3) NOR

(4) NAND

Answer Key

Q1 (3)	**Q2** (4)	**Q3** (2)	**Q4** (3)
Q5 (3)	**Q6** (20)	**Q7** (2)	**Q8** (5)
Q9 (4)	**Q10** (120)	**Q11** (4)	**Q12** (1)
Q13 (3)			

Q1 (3)

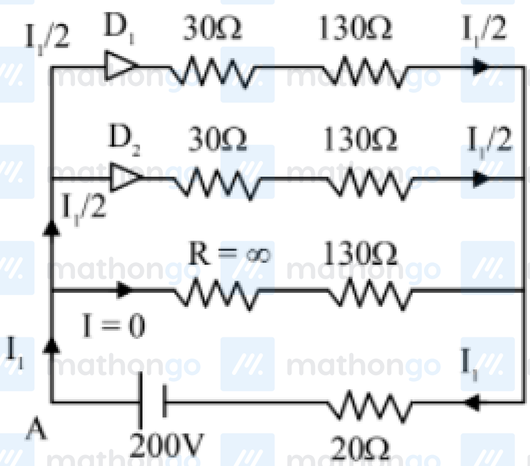


As per diagram,

Diode D_1&D_2 are in forward bias i.e. $R = 30\Omega$ whereas diode D_3 is in reverse bias i.e. $R =$ infinite $\Rightarrow$

Equivalent circuit will be

Applying KVL starting from point A



$$-\left(\frac{I_1}{2}\right) \times 30 - \left(\frac{I_1}{2}\right) \times 130 - I_1 \times 20 + 200 = 0$$
$$\Rightarrow -100 I_1 + 200 = 0$$
$$I_1 = 2$$

Option (3)

Q2 (4)

To convert pulsating dc into steady de both of mentioned method are correct.

Q3 (2)

$$\begin{aligned} V_A = 5\text{ V} &\Rightarrow A = 1 \\ V_A = 0\text{ V} &\Rightarrow A = 0 \\ V_B = 5\text{ V} &\Rightarrow B = 1 \\ V_B = 0\text{ V} &\Rightarrow B = 0 \end{aligned}$$

If $A = B = 0$, there is no potential anywhere here $V_0 = 0$

If $A = 1, B = 0$, Diode D_1 is forward biased, here $V_0 = 5$ V

If $A = 0$, $B = 1$, Diode D_2 is forward biased hence $V_0 = 5$ V

If $A = 1$, $B = 1$, Both diodes are forward biased hence $V_0 = 5$ V

Truth table for I^{st}

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

$\therefore$ Given circuit is OR gate

For II^{nd} circuit

$$\begin{aligned} V_B = 5\text{ V}, &\quad A = 1 \\ V_B = 0\text{ V}, &\quad A = 0 \end{aligned}$$

When $A = 0, E - B$ junction is unbiased there is no current through it

$$\therefore V_0 = 1$$

When $A = 1, E - B$ junction is forward biased

$$V_0 = 0$$

$\therefore$ Hence this circuit is not gate.

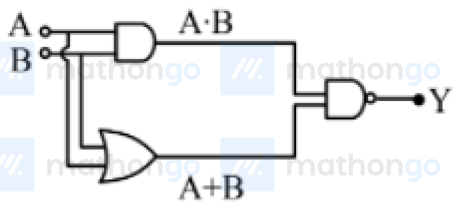
Q4 (3)

When $V_i > 3$ volt, $V_R > 0$
Because diode will be in forward biased state

When $V_i \leq 3$ volt ; $V_R = 0$

Because diode will be in reverse biased state.

Q5 (3)



$$Y = \overline{(A \cdot B) \cdot (A + B)}$$

$\left.Y\right)_{(0,0)} = 1$

$\left.Y\right)_{(0,1)} = 1$

$\left.Y\right)_{(1,0)} = 1$

$\left.Y\right)_{(1,1)} = 0$

Option (3) is correct

Q6 (20)

When unregulated voltage is 14 V voltage across zener diode must be 10 V So potential difference across resistor $\Delta V_{Rs} = 4$ V

and $P_{zener} = 2$ W

$VI = 2$

$I = \frac{2}{10} = 0.2 \text{ A}$

$\Delta V_{R_s} = IR_s$

$4 \times 0.2R_s \Rightarrow R_s = \frac{40}{2} = 20\Omega$

Q7 (2)

$\alpha = \frac{I_C}{I_E}, \beta = \frac{I_C}{I_B}; I_E = I_C + I_B$

$\alpha = \frac{I_C}{I_C + I_B} = \frac{I_C/I_B}{\frac{I_C}{I_B}+1} = \frac{\beta}{\beta+1} +$

$1 + \frac{1}{\beta} = \frac{1}{\alpha}$

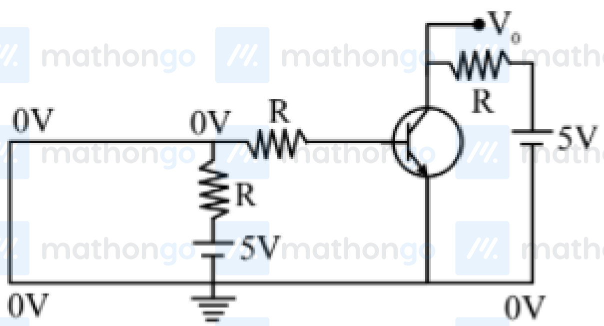
$\frac{1}{\beta} = \frac{1-\alpha}{\alpha}$

$\beta = \frac{\alpha}{1-\alpha}$

Q8 (5)

As diodes D_1 and D_2 are in forward bias, so they acted as neligible resistances

$\Rightarrow$ Input voltage become zero



$\Rightarrow$ Input current is zero

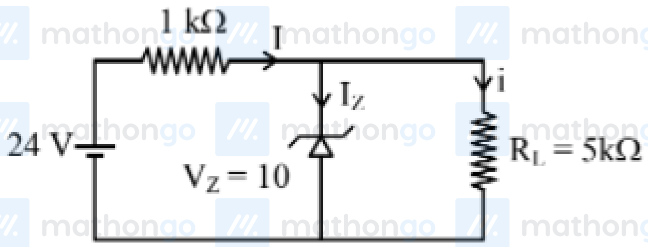
$\Rightarrow$ Output current is zero

$\Rightarrow V_0 = 5$ volt

Q9 (4)

Active region of the CE transistor is linear region and is best suited for its use as an amplifier

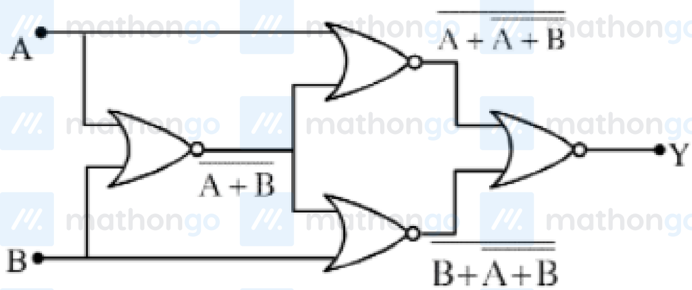
Q10 (120)



$$i = \frac{10\text{ V}}{5k\Omega} = 2\text{ m A}$$
$$I = \frac{14\text{ V}}{1k\Omega} = 14\text{ mA}$$
$$\therefore I_z = 12\text{ mA}$$
$$\therefore P = I_z V_z = 120\text{ mW}$$

Ans. 120

Q11 (4)



$$y = \overline{\overline{(A + A + B)} + \overline{(B + A + B)}}$$
$$y = (A + \overline{A + B}) \cdot (B + \overline{A + B})$$

A	B	y
0	0	1
0	1	0
1	0	0
1	1	1

Ans.4

Q12 (1)

Pentavalent activities have excess free e^-

So e^-density increases but overall semiconductor is neutral.

Option (1)

Q13 (3)

A	B	X	Y	Z
1	1	0	0	0
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

Option (3)